

Solutions to Combinatorics Problems

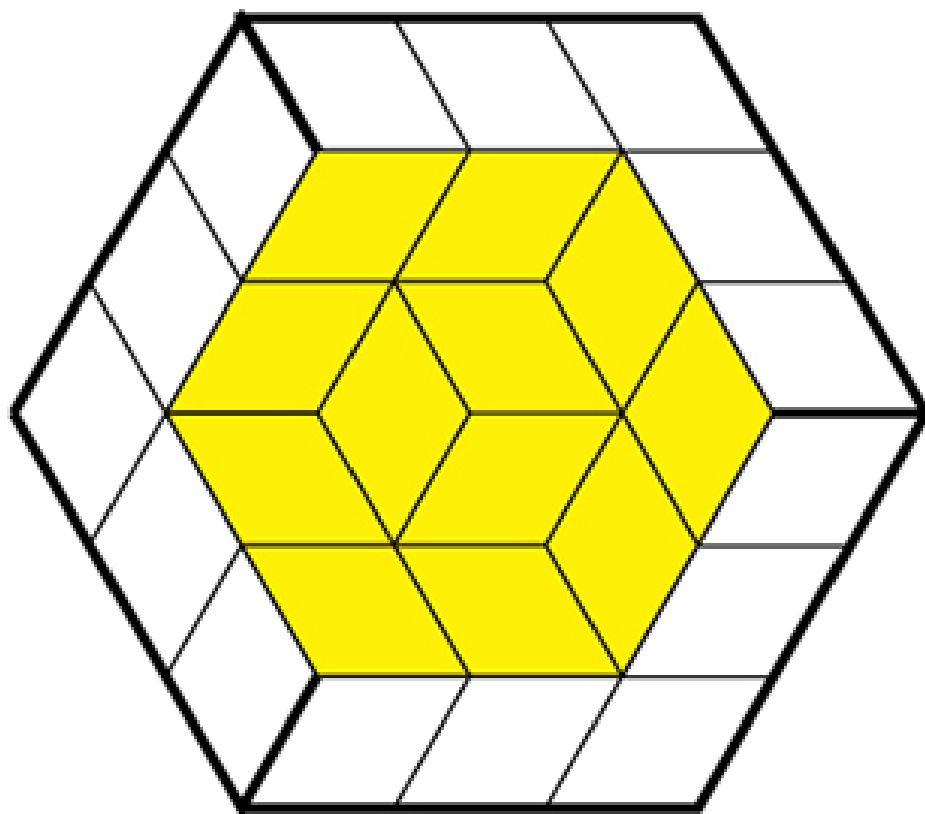
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Abstract

This note collects my solutions to selected problems in Combinatorics. The goal is to present clear arguments and organized solutions for study and review. Books that I use for assignments sources include:

- A Walk Through Combinatorics, Miklós Bóna, 5ed



Problem 1.

For which values of n does there exist a balanced tournament on n vertices?

Source: *A Walk Through Combinatorics - Quick check 2, p.221*

Solution:

We claim that there exists a balanced tournament on n vertices if and only if n is odd.

($\Rightarrow$): Since for every vertex v , we have $\text{indeg}(v) = \text{outdeg}(v) = \frac{n-1}{2}$, n must be odd.

($\Leftarrow$): Suppose we have an odd number of vertices (n is odd). We will prove by induction that there exists a balanced tournament G on n vertices, $n = 2k + 1$, $k \in \mathbb{N}$.

- We can easily see that $k = 0$ and $k = 1$ is true.
- Now suppose $P(k)$ is true for some $k \in \mathbb{N}$, we will show that $P(k + 1)$ is true.

Let G be a balanced tournament over $n = 2k + 1$ vertices. Consider two additional vertices v_1 and v_2 . We need to create a balanced tournament $\tilde{G}$ by connecting v_1 and v_2 to G . This can be done by:

1. Create an edge $v_1 \rightarrow v_2$.
2. Create k edges from v_1 to k vertices in G . (set U)
Create $k + 1$ edges from $k + 1$ remaining vertices in G to v_1 . (set W)
3. Create k edges from set U to v_2 .
Create $k + 1$ edges from v_2 to W .

By induction, this is true for all $k \in \mathbb{N}$.

Problem 2.

Does there exist a tournament with exactly two Hamiltonian paths?

Source: *A Walk Through Combinatorics - Quick check 3, p.221*

Solution:

Let G be a tournament. Suppose G is strongly connected. Then G has a Hamiltonian cycle by Theorem 9.8. Therefore, G can never have exactly 2 Hamiltonian paths. (1)

Now suppose G is not strongly connected. We will group vertices of G into strongly connected components $C_1, C_2, \dots, C_k$ ($k \geq 2$). This can be done because a group of vertices in the same strongly connected component forms an equivalence class.

Every edge from a vertex in C_i to a vertex in C_j must also point in the same direction. Otherwise, from any vertex in C_i , we can go to any vertex in C_j and vice versa. Also, if $C_i \rightarrow C_j$ and $C_j \rightarrow C_k$, then $C_i \rightarrow C_k$.

Because of this transitive property, the components can be listed from left to right so that $C_i \rightarrow C_j$ if and only if C_j is on the right of C_i . Let's label this sequence from left to right as $C_1, C_2, \dots, C_k$. Let $H(G)$ denote the number of Hamiltonian paths in a tournament G . We have:

$$H(G) = H(C_1) \times H(C_2) \times \dots \times H(C_k)$$

Note that each C_i considered on its own is also a tournament. For $H(G) = 2$, exactly one of the C_i 's must have $H(C_i) = 2$, and for all other C_j 's, $H(C_j) = 1$. But this is impossible by (1).

Therefore, there exist no tournaments with exactly 2 Hamiltonian paths.

Problem 3.

Let G be a loopless undirected graph. Prove that the edges of G can be directed so that no directed cycle is formed.

Source: A Walk Through Combinatorics - Ch.9 Ex.1 p.225

Solution:

Let G be a graph on n vertices. We need to show the edges of G can be directed in such a way that no directed cycle is formed. Let's label vertices from 1 to n .

First, make every of $n - 1$ edges connected to 1 be an out edge. It follows that no possible directed cycle can start from 1.

Now, consider all remaining $n - 2$ undirected edges connected to 2 and make each of them to be an out edge. It also follows that no directed cycle can start from 2.

Similarly, continue with this labelling scheme, no directed cycle can start from any of the vertices.

Problem 4.

Let G be a simple graph on 10 vertices and 28 edges. Prove that G contains a cycle of length 4.

Source: A Walk Through Combinatorics - Ch.9 Ex.3 p.225

Solution:

It seems like we are having too many edges so that we can always form a cycle of length 4. Saying that there always exists a cycle of length 4 is equivalent to saying that there always exists two vertices A and B such that the intersection of their set of neighbors has at least two vertices. In other words, we want to find two vertices A and B such that there are at least two or more vertices that connect to both A and B.

So why there always exists such two vertices A and B? The intuition here is that if A connects to

a lot of neighbors and B also has a lot of neighbors then it's likely that we can find at least two vertices connecting to both. So let A and B be two vertices with the highest degrees. Since G has 28 edges, sum of all degrees of vertices = 56. It follows that $\deg(A) + \deg(B) \geq 12$. Even if there is an edge between A and B, there are still at least 10 edges between other 8 points and A or B. This implies at least two or more vertices that connect to both A and B.

Problem 5.

Let G be a graph. Recall that we say that H is an *induced subgraph* of G if the vertex set of H is a subset of that of G , and if x and y are two vertices of H , then xy is an edge in H if and only if xy is an edge in G .

Let G be a simple graph that has 10 vertices and 38 edges. Prove that G contains K_4 (the complete graph on four vertices) as an induced subgraph.

Source: A Walk Through Combinatorics - Ch.9 Ex.5 p.225

Solution: